

Kernel-Based Survival Methods for Estimating Returns to Female Schooling

A Nonparametric Approach to Educational Trajectories

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Outline

- 1 Motivation
- 2 Contribution
- 3 Empirical Strategy & Results
- 4 Conclusions & Broader Implications

Why Does Female Education Matter?

Well-Documented Returns to Education

- Education improves welfare broadly (Duflo, Dupas, and Kremer 2015)
- Historically, differential treatment has led to lower rates of early school enrollment among female students (Duflo 2011)

Macro-Level Channels

- Improved fund of knowledge $\Rightarrow$ economic growth (Hanushek 2009)
- Higher enrollment $\Rightarrow$ expanded female labour supply (Bloom et al. 2009)

Prior studies used simple models and few variables and had been constrained by computational resources and data size, limiting predictive power for policy analysis.

Limitations of the Existing Literature

- Returns to education are treated as **linear**, as in Manski's original formulation (Card 1999):
 - ① Separability between effects of education and experience
 - ② Linearity of standard education models
- But linearity need not hold for all populations at all times:
 - “Sheepskin” (diploma) effects (Card 1999)
 - Heterogeneity across populations (Koop and Tobias 2002; Aakvik, Salvanes, and Vaage 2010)
 - Prior work captures specific nonlinearities but still assumes the standard Mincer equation (Hungerford and Solon 1987; Belman and Heywood 1991)
- **Goal:** a flexible method to make good predictions, inform broader inference, and support better-tailored policies

This Paper's Contribution

Novel Methodology

We measure returns to education via a **survival function using kernel estimation** — relaxing linearity and parametric assumptions.

Advantage 1: Survival Modelling

- More precise than standard OLS (Wu, Dharamshi, and Wakefield 2025)
- Estimates hazard (dropout probability) and survival (retention probability) functions
- Corrects for right-censoring bias

Advantage 2: Kernel Methods

- Fully nonparametric — no linearity restriction (Manski and Pepper 2000; Ulrick 2007)
- Reduces omitted variable bias
- Handles high-dimensional covariates
- No multicollinearity issues

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This is the first paper to apply **kernel-based survival estimation** to policy analysis for girls' educational attainment.

Why Survival Methods for Girls' Education?

Censoring Problem

Female students are typically **right-censored** in the data:

- At the end of monitoring, we observe that some girls are still enrolled
- But we want to know: *how long will they stay, and what is the probability they remain enrolled in a given year?*

What Survival Methods Provide

- **Hazard function:** probability of dropping out at a specific time t
- **Survival function:** probability of remaining enrolled up to time t
- A dynamic, bias-corrected, spatially disaggregated measure of attainment (Wu, Dharamshi, and Wakefield 2025)

Why Kernel Methods over Cox Regression?

Cox Model Limitations

- Assumes *constant* hazard ratio over time
- Does not necessarily apply where selection bias or time-varying treatment create identification issues
- Struggles with multicollinearity
- Sensitive to high-dimensional covariates

Kernel Method Advantages

- **No parametric restrictions**
- Captures complex, non-linear earnings profiles (Heckman, Lochner, and Todd 2006)
- Accommodates sheepskin effects and population heterogeneity
- No multicollinearity or dimensionality issues

There is little justification for linearity when families face the choice of keeping daughters in school — especially given well-documented heterogeneity in educational returns.

Data and Empirical Strategy

- **Data:** Cross-sectional panel data on educational treatment across countries
- **Estimation:** Survey-weighted logistic regression, calibrated to national data
- **Method:** Kernel-based survival estimation applied to longitudinal studies
- **Validation:** Check model against other benchmark econometric methods
- **Extension:** Fine-tuned model for counterfactual policy predictions?

Roadmap to Results

- 1 Validate model accuracy
- 2 Estimate female educational trajectories corrected for censoring and omitted variable bias
- 3 Identify groupings of variables and interactions significant for keeping girls in school
- 4 Generate counterfactual predictions for potential policy actions
- 5 Characterize channels through which educational interventions operate ¹

¹(Vafa, Athey, and Blei 2024; Chen 2022)

Conclusions & Broader Implications

Summary

- Kernel-based survival methods offer a flexible, bias-corrected approach to estimating returns to girls' schooling (especially applicable to LMIC)
- First application of this methodology to girls' educational policy analysis

Broader Applicability

This framework extends beyond education:

- Any attrition or time-to-event policy problem
- Policy analysis and identification of empirical groupings or factors affecting probability of graduation or time to graduation
- Kernel methods on panel data as a general tool — adding structure and interpretability to machine learning in policy evaluation

References I

-  Aakvik, Arild, Kjell G Salvanes, and Kjell Vaage (2010). "Measuring heterogeneity in the returns to education using an education reform". In: *European Economic Review* 54.4, pp. 483–500.
-  Belman, Dale and John S. Heywood (1991). "Sheepskin Effects in the Returns to Education: An Examination of Women and Minorities". In: *The Review of Economics and Statistics* 73.4, pp. 720–724. ISSN: 0034-6535. DOI: 10.2307/2109413. URL: <https://www.jstor.org/stable/2109413> (visited on 04/30/2026).
-  Bloom, David E. et al. (June 1, 2009). "Fertility, female labor force participation, and the demographic dividend". In: *Journal of Economic Growth* 14.2, pp. 79–101. ISSN: 1573-7020. DOI: 10.1007/s10887-009-9039-9. URL: <https://doi.org/10.1007/s10887-009-9039-9> (visited on 04/30/2026).

References II



Card, David (1999). “The Causal Effect of Education on Earnings”. In: *Handbook of Labor Economics*. Vol. 3. Elsevier, pp. 1801–1863. ISBN: 978-0-444-50187-5. DOI: 10.1016/S1573-4463(99)03011-4. URL: <https://linkinghub.elsevier.com/retrieve/pii/S1573446399030114> (visited on 04/27/2026).



Chen, George H. (June 21, 2022). *Survival Kernets: Scalable and Interpretable Deep Kernel Survival Analysis with an Accuracy Guarantee*. arXiv.org. URL: <https://arxiv.org/abs/2206.10477v6> (visited on 04/29/2026).



Duflo, Esther (2011). *Women’s Empowerment and Economic Development*. w17702. Cambridge, MA: National Bureau of Economic Research, w17702. DOI: 10.3386/w17702. URL: <http://www.nber.org/papers/w17702.pdf> (visited on 04/30/2026).

References III

-  Duflo, Esther, Pascaline Dupas, and Michael Kremer (Sept. 2015). “Education, HIV, and Early Fertility: Experimental Evidence from Kenya”. In: *American Economic Review* 105.9, pp. 2757–2797. ISSN: 0002-8282. DOI: 10.1257/aer.20121607. URL: <https://www.aeaweb.org/articles?id=10.1257/aer.20121607> (visited on 04/27/2026).
-  Hanushek, Eric A (2009). “The Economic Value of Education and Cognitive Skills”. In: *Handbook of Education Policy Research*, pp. 39–56.
-  Heckman, James Joseph, Lance J Lochner, and Petra E Todd (2006). “Earnings functions, rates of return and treatment effects: the Mincer equation and beyond”. In.
-  Hungerford, Thomas and Gary Solon (1987). “Sheepskin Effects in the Returns to Education”. In: *The Review of Economics and Statistics* 69.1, pp. 175–177. ISSN: 0034-6535. DOI: 10.2307/1937919. URL: <https://www.jstor.org/stable/1937919> (visited on 04/30/2026).

References IV

-  Koop, Gary and Justin L Tobias (Mar. 2002). “Learning About Unobserved Heterogeneity in Returns to Schooling”. In.
-  Manski, Charles F. and John V. Pepper (2000). “Monotone Instrumental Variables: With an Application to the Returns to Schooling”. In: *Econometrica* 68.4, pp. 997–1010. ISSN: 0012-9682. URL: <https://www.jstor.org/stable/2999533> (visited on 04/30/2026).
-  Ulrick, Shawn W. (Oct. 1, 2007). “Measuring the returns to education nonparametrically”. In: *Applied Economics Letters* 14.13. eprint: <https://doi.org/10.1080/13504850600706198>, pp. 1005–1011. ISSN: 1350-4851. DOI: 10.1080/13504850600706198. URL: <https://doi.org/10.1080/13504850600706198> (visited on 04/30/2026).
-  Vafa, Keyon, Susan Athey, and David M. Blei (Sept. 15, 2024). *Estimating Wage Disparities Using Foundation Models*. arXiv.org. URL: <https://arxiv.org/abs/2409.09894v2> (visited on 04/30/2026).



Wu, Yunhan, Ameer Dharamshi, and Jon Wakefield (Feb. 11, 2025). *Small Area Estimation of Education Levels in Low- and Middle-Income Countries*. DOI: 10.48550/arXiv.2502.07946. arXiv: 2502.07946[stat]. URL: <http://arxiv.org/abs/2502.07946> (visited on 04/29/2026).